

simulate static routing Configuration using Cisco Packet Tracer.

Aim :

To simulate a static routing Configuration using Cisco Packet tracer.

Procedure :

- ★ Open Cisco Packet tracer and Create the given network topology.
- ★ Connect 3 routers and PC's using suitable cable.
- ★ Configure IP address for routers and PCs.
- ★ Enable all routers interface using no shutdown.
- ★ Configure static router on Router 0; router 1, router 2 using ip mobile.
- ★ Set main & backup router using different administrative distance values.
- ★ Test Connectivity.
- ★ Use tracer to verify the routing path.
- ★ Disconnect the main link and verify that the backup router works.
- ★ Thus a static routing is successfully Configured and verified.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  40.0.0.1        YES manual up          up
GigabitEthernet0/1  10.0.0.1        YES manual up          up
GigabitEthernet0/2  20.0.0.1        YES manual up          up

Router1#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
              B - BGP, D - EIGRP, EX - EIGRP external

Gateway of last resort is not set

S    30.0.0.0/8 via 20.0.0.2
S    10.0.0.0/8 via 10.0.0.2
Router1#
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  40.0.0.2        YES manual up          up
GigabitEthernet0/1  20.0.0.2        YES manual up          up

Router1#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
              B - BGP, D - EIGRP, EX - EIGRP external

Gateway of last resort is not set

S    10.0.0.0/8 via 20.0.0.1
S    10.0.0.0/8 via 20.0.0.1
Router1#
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  40.0.0.2        YES manual up          up
GigabitEthernet0/1  30.0.0.1        YES manual up          up

Router2#show ip route static
Static Codes: S - static, I - IGRP, B - BGP, M - mobile,
              R - RIP, D - EIGRP, EX - EIGRP external

Gateway of last resort is not set

S    20.0.0.0/8 via 40.0.0.1
S    10.0.0.0/8 via 40.0.0.1
Router2#
```

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  30.0.0.1        YES manual up          up

Router3#show ip route static
Static Codes: S - static, I - IGRP, B - BGP, M - mobile,
              R - RIP, D - EIGRP, EX - EIGRP external

Gateway of last resort is not set

S    10.0.0.0/8 via 30.0.0.2
S    20.0.0.0/8 via 30.0.0.2
Router3#
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 30.0.0.2
Pinging 30.0.0.2 with 32 bytes of data:
Reply from 30.0.0.2: bytes=32 time=1ms TTL=127
Reply from 30.0.0.2: bytes=32 time=1ms TTL=127
Reply from 30.0.0.2: bytes=32 time=1ms TTL=127
Reply from 30.0.0.2: bytes=32 time=1ms TTL=127
Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 30.0.0.100
Pinging 30.0.0.100 with 32 bytes of data:
Reply from 30.0.0.100: bytes=32 time=1ms TTL=127
Reply from 30.0.0.100: bytes=32 time=1ms TTL=127
Reply from 30.0.0.100: bytes=32 time=1ms TTL=127
Reply from 30.0.0.100: bytes=32 time=1ms TTL=127
Ping statistics for 30.0.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time=1ms TTL=127
Reply from 10.0.0.2: bytes=32 time=1ms TTL=127
Reply from 10.0.0.2: bytes=32 time=1ms TTL=127
Reply from 10.0.0.2: bytes=32 time=1ms TTL=127
Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

OK



all links are up

Ex No 116
20/9/26

Simulate RIP using CISCO PACKET TRACER.

Aim:

To simulate RIP using Cisco Packet Tracer.

Procedure:

- * Create the topology with 3 routers and 2 PCs.
- * Connect the devices using suitable cables.
- * Assign the given IP address to both PCs.
- * Configure Fast Ethernet and serial interface of all routers.
- * Enable interface using router rip.
- * Add directly connected networks using the network command.
- * Test Connectivity using ping 20.0.0.2
- * Use traces 20.0.0.2 to verify the selected router.
- * Disconnect the direct Router 0 - Router 2 link and use tracer again to verify alternate route. ~~RIP~~ selects the

PC0 - Command Prompt (ping)

```
C:\> ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\> tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  0  1 ms    1 ms    1 ms   10.0.0.1
  1  2 ms    2 ms    2 ms   192.168.1.253
  2  3 ms    3 ms    3 ms   20.0.0.2

Trace complete.

C:\>
```

Slt

Result:

Thus
TRACER

RIP

was

using
implemented

CISCO

PACKET

successfully.